Hyperparameter tuning results

Best tested settings, selected on validation data and tuning timings before confirmation/test evaluation. Single training seed; speed and quality claims are limited to this grid, workload, GPU, and FA1-derived prototype.

All timings use FP32 stored weights with FP16 autocast, batch 1, prompt 896, 64 generated tokens. LoRA is unmerged. Cache tuning changes prefill chunk size without evicting tokens. FA tuning changes query/key tile sizes and warp count; every FA configuration uses the FA1-derived algorithm, not the original optimized FA1 CUDA library.

Selected LoRA: {'rank': 2, 'alpha': 4, 'lr': 0.001}. Selected cache prefill chunk: 896. Selected FA tiles/warps: [32, 64, 4].

Comparison	Untuned ms	Tuned ms	Speedup
lora	1107.943	1106.663	1.001×
cache	878.096	797.850	1.101×
fa	1510.178	1077.882	1.401×
combined	819.565	743.967	1.102×
lora_training_step	65.950	66.878	0.986×

Ratios below 1 are slowdowns. The predeclared untuned controls may themselves win; we do not deliberately choose the slowest setting. LoRA inference and training speedups are reported separately. KV cache has no role in the training step.

The combined row holds all three methods enabled and compares the individual winners against rank 4 / alpha 8 / LR 0.0001, cache prefill chunk 128, and FA tiles 16×32 / 4 warps. Individual winners are not jointly reoptimized. Confirmation uses fresh prompts, seven shuffled paired repeats, and bootstrap intervals. Prompt length, batch size, output length, precision and cache retention are held fixed.

LoRA eligibility requires validation NLL within 0.02 of the best grid candidate; eligible candidates are ranked by tuning inference latency. The untuned LoRA control is reported even if it fails that quality constraint. Runtime settings must pass tensor correctness and full validation NLL checks. FP16 chunk sizes can change rounding and greedy tokens; cache settings are gated on same-input logit closeness and validation NLL, and raw token differences are preserved. Test data does not select any settings.

Model	Validation NLL	Test fields accuracy
baseline	0.07581	0.500
tuned	0.02535	0.825

Raw timing samples, token outputs, source/data hashes, frozen selections, quality scores and checkpoints are preserved beside this report. Training/search overhead is not included in inference latency. There is no claim that tuning amortizes its search cost.

Confirmation uncertainty

Paired bootstrap 95% intervals describe timing variability within this allocation; they do not measure uncertainty across hardware or training seeds.

Comparison	Speedup	95% interval
lora	1.001×	1.001–1.002×
cache	1.101×	1.098–1.103×
fa	1.401×	1.401–1.401×
combined	1.102×	1.094–1.105×

All LoRA candidates

Rank	Alpha	LR	Validation NLL	Selected step	Training ms/step	Eligible
2	2	0.0001	0.09127	1200	66.930	False
2	2	0.001	0.04773	1200	64.556	False
2	4	0.0001	0.08631	1200	66.906	False
2	4	0.001	0.02527	1200	66.878	True
4	4	0.0001	0.08520	900	67.316	False
4	4	0.001	0.05190	1200	66.254	False
4	8	0.0001	0.07592	900	65.950	False
4	8	0.001	0.02801	1200	67.188	True

Runtime tuning timings

These are selection timings; final speedups above use fresh confirmation measurements.

Method	Setting	Median ms
lora	{'rank': 2, 'alpha': 4, 'lr': 0.001} (eligible=True)	1108.067
lora	{'rank': 4, 'alpha': 8, 'lr': 0.0001} (eligible=False)	1108.758
lora	{'rank': 4, 'alpha': 8, 'lr': 0.001} (eligible=True)	1108.835
cache	64	951.701
cache	128	857.125
cache	256	816.471
cache	896	777.055
fa	(16, 32, 4)	1511.592
fa	(16, 32, 8)	1563.642
fa	(16, 64, 4)	1146.654
fa	(16, 64, 8)	1247.897
fa	(32, 32, 4)	1295.980
fa	(32, 32, 8)	1432.571
fa	(32, 64, 4)	1078.616
fa	(32, 64, 8)	1123.675

Cache and FA sweeps and their individual confirmation comparisons hold the untuned LoRA checkpoint fixed. LoRA-only confirmation uses eager attention with caching disabled. The combined comparison changes all three selected settings together.

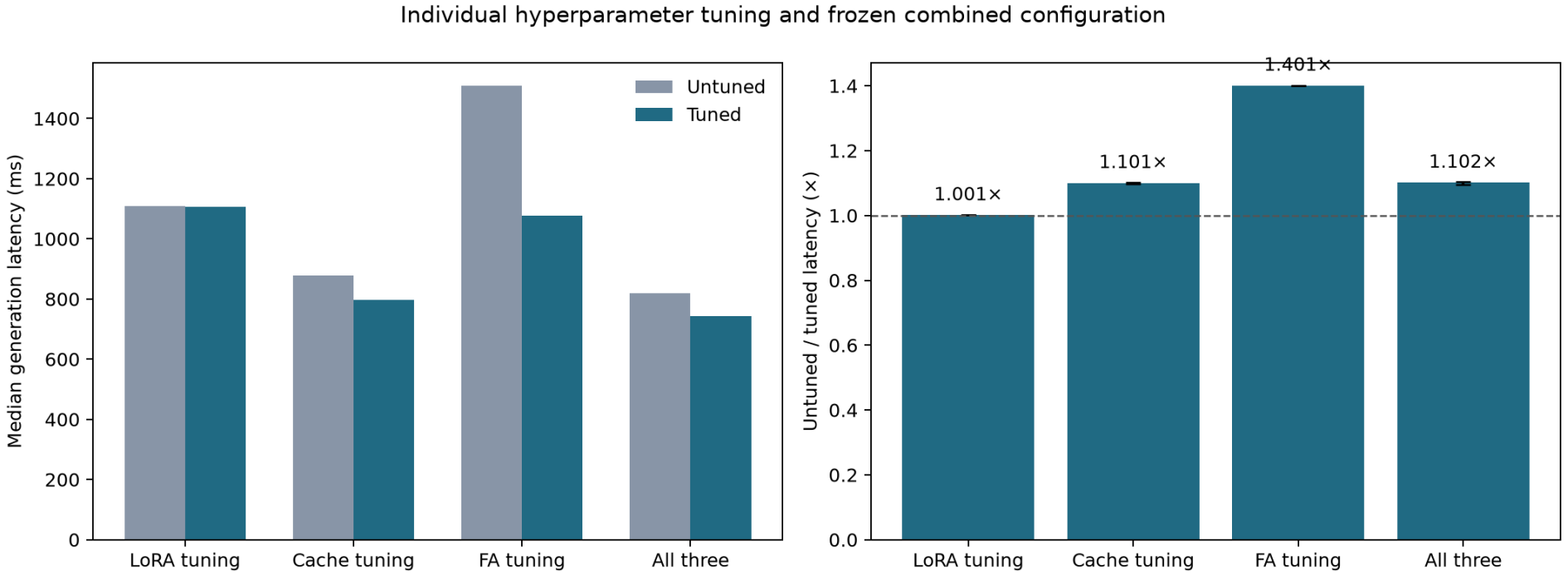
Cache numerical checks and protocol amendment

The initial runtime sweep stopped on an exact greedy-token check. Before confirmation and test evaluation, the recorded protocol was amended to require same-input logit closeness and validation NLL parity; all eight training runs were reused. FP16 execution shapes affect rounding. This amendment and the observed divergence are part of the result.

Prefill chunk	Validation NLL	Max last-prefill logit error	Greedy tokens equal full prefill
64	0.075571	0.09375	True
128	0.075923	0.06250	True
256	0.075923	0.06250	False
896	0.075923	0.00000	True

Held-out combined answer quality

Configuration	Primary correct	Numbers correct	Fields correct	Exact reference
baseline	0.500	1.000	0.500	0.500
tuned	0.825	1.000	0.825	0.825


Individual hyperparameter tuning and frozen combined configuration
Median generation latency (ms)
Untuned
Tuned
0
200
400
600
800
1000
1200
1400
LoRA tuning
Cache tuning
FA tuning
All three
Untuned / tuned latency (×)
1.001×
1.101×
1.401×
1.102×
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
LoRA tuning
Cache tuning
FA tuning
All three